Jinghua Xi

Bonn, Germany | s79jxi@uni-bonn.de
jinghua-xi.github.io

Mathematical Background and Skills

Core areas: Partial differential equations, functional analysis, and probability theory.
Programming: Python; project experience with JavaScript and HTML/CSS.
Tools used: Git, SQLite, REST APIs and Ollama.

Education

University of Bonn Oct 2025 – Present
M.Sc. in Mathematics Bonn, Germany

- **Current overall GPA: 1.0** (German grading scale; 1.0 is the highest grade).
- **Analysis and PDE:** PDE and Functional Analysis (1.0); PDE and Modelling (1.0).
- **Probability:** Stochastic Processes (1.3); Optimal Transport Seminar (1.0).

Southern University of Science and Technology Sep 2021 – Jun 2025
B.Sc. in Mathematics, Fields Honors Program Shenzhen, China

Publications

[1] C. G. Moreira, J. Xi and Y. Zhang (2025): **Graphs of continuous but non-affine functions are never self-similar.** *Proc. Amer. Math. Soc.* 153, 2501–2512. doi:10.1090/proc/17099.

Main result: Proved that the graph of a continuous function on $[0, 1]$ is self-similar if and only if it is a straight line.

Selected Projects

AI-Assisted B2B Ordering Prototype Nov 2026

- Developed and deployed a Python/SQLite ordering prototype integrating self-hosted Qwen3-8B through Ollama on a cloud GPU.
- Built retrieval over a 100-SKU catalogue, procurement-text parsing, multi-item carts, role-based access and Excel/email workflows.
- Validated 52 regression tests and implemented one-command SSH deployment with backups and health checks. Project

LLM Deployment and GPU Inference Nov 2026
Deployed 4-bit quantized Qwen3-8B on an NVIDIA L4 GPU in Google Colab using llama.cpp and CUDA. Used Python to validate an OpenAI-compatible REST API for LLM inference. Project

Honors and Awards

Rixin Motto Scholarship Oct 2024
Southern University of Science and Technology Details

- Awarded to 9 students for academic and overall excellence.

Exchange and Summer Schools

[Exchange] Enhanced Program for Graduate Study Feb 2024 – Jun 2024
Peking University, China Program

- Selected as one of 31 participants nationwide.

[Summer School] Geometry Aug 2023
Utrecht University, Netherlands Program

[Summer School] Mathematics of Random Systems Aug 2024
University of Oxford, United Kingdom Program

[Summer School] Maximal Functions and Differentiation of Integrals in Euclidean Space Jun 2025
Westlake University, China Program

Leadership Experience

Founder & Organizer, Undergraduate Mathematics Salon Sep 2023 – Jun 2025
Southern University of Science and Technology Archive

- Founded a seminar series promoting clear mathematical exposition and discussion among undergraduates.
- Coordinated monthly events and encouraged students to present and discuss mathematical topics.
- Featured on the university's official WeChat account. Read more.